

RITHAN KOUSHIK K

rithankoushik05@gmail.com | [GitHub](#) | [LinkedIn](#) | [LeetCode](#) |

+91-7200722001 | Chennai, India

Executive Summary

Software Engineer with strong foundations in data structures, algorithms, and backend development, complemented by hands-on experience in AI/ML systems. Experienced in building scalable applications, REST APIs, and data-driven solutions using Python, SQL, and FastAPI. Worked on production-grade systems involving cloud-native deployment, containerization, and model pipelines, with industry experience at Zoho.

Skills

- **Programming & Core CS:** Python, SQL, Data Structures & Algorithms, OOPS
- **ML & DL:** PyTorch, Scikit-learn, Transformers, NLP, LLM Fine-tuning, LangChain, AI Agents, Ollama.
- **MLOps & DevOps:** Docker, Kubernetes, MLflow, Jenkins (CI/CD), Git, Linux (Ubuntu)

Internship Experience

AI/ML Engineer Intern at Sacha Global Jan 2026 – Present

- Built an **AI-powered Invoice Processing System** using **OCR and LLM** to extract structured invoice data and generate **validated JSON outputs**. Automated **SharePoint list auto-fill** by mapping LLM-generated JSON fields to enterprise forms, reducing manual data entry and improving document processing accuracy.
- Developed an **unsupervised vehicle segmentation system** using **clustering** on vehicle technical specifications to classify economy, mid-range, and luxury segments.
- Built a **YOLOv8**-based vehicle damage detection system to accurately detect and localize damaged regions, enabling automated inspection and insurance workflows

Summer Intern at Zoho Corporation June 2025 – Nov 2025

- **Fine-tuned Transformer-based models (Qwen3, T5)** for NLP tasks, building a **supervised job-description parsing system** that converted unstructured JDs into structured JSON for CRM automation. Contributed to evaluation pipelines for **RAG and vision-language models**, improving response relevance and consistency.
- **Fine-tuned Wav2Vec2 on the MINDs14 dataset for voice command recognition**, achieving improved intent classification through domain-specific adaptation. Integrated **ML workflows into Zoho Agent Studio**, enabling scalable deployment and monitoring of **AI agents**. [project-link](#)

Data Analyst Intern at Vyastics Technology Jan 2025 – Apr 2025

- Created a healthcare dashboard using Tableau with USA hospital data, featuring state-wise hospital ratings and KPIs like readmission rate, mortality rate, emergency services, and overall hospital counts. [project-link](#)

Education & Certifications

- **Bachelor of Technology (Artificial Intelligence and Data Science)**
Rajalakshmi Institute of Technology, Chennai, *CGPA: 8.2* 2022–Present
- **12th Grade** – Sri Venkateswara Matriculation Higher Secondary School, *83%* 2021–2022
- **AI Engineer for Data Scientists Associate** — [DataCamp -link](#)

Projects

Fit_AI – AI-Powered Fitness & Nutrition Assistant [project-link](#)

- Built a full-stack AI fitness application that classifies body type and **predicts user height, weight, and BMI from full-body images using a fine-tuned Vision Transformer (ViT) model**. Generated personalized workout and diet plans using **LLMs via the Groq API**, and implemented **multilingual voice-enabled AI chatbot support (100+ languages)** for natural user interaction.
- Integrated **JWT-based authentication**, a **diet tracking system with real-time macro calculations** via the Edamam API, and delivered a responsive modern UI using **React and TailwindCSS**, backed by a **FastAPI + MongoDB** backend.

Multimodal-RAG AI-Tutor [project-link](#)

- Built a **multimodal Retrieval-Augmented Generation (RAG) chatbot** enabling **text and image-based document querying**, supporting PDFs and visual inputs with context-aware AI responses.
- Implemented end-to-end AI pipelines using **Streamlit, Gemini AI, BLIP, ChromaDB, LangChain, VectorDB and SentenceTransformers** for document processing, embeddings generation, semantic retrieval, and response

CO2 Emission Prediction System for Automobiles [project-link](#)

- Designed and deployed a supervised regression pipeline(**Random Forest model**) to estimate vehicle CO2 emissions from automotive specifications, including data preprocessing, feature engineering, model comparison, and real-time inference via FastAPI, with a Streamlit frontend for user interaction.